



Technical Specification 43-50

Issue 2 2012

SPECIFICATION FOR SINGLE CIRCUIT
OVERHEAD LINES ON WOOD POLES FOR USE
AT 132kV

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**Operations Directorate
Energy Networks Association
6th Floor, Dean Bradley House
52 Horseferry Rd
London
SW1P 2AF**

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SPECIFICATION FOR SINGLE CIRCUIT OVERHEAD LINES ON WOOD POLES FOR USE AT 132kV

Foreword

Issue 1 of ENATS 43-50 was originally produced in 1984. The introduction of EN50341 Overhead Electrical Lines Exceeding AC 45kV has produced the need to review this Specification. This has given the opportunity to move the design process on from the paper based process given in Issue 1 to the spreadsheet based process presented in this Issue.

1. Scope

EN50341 gives the design requirements for overhead lines operating at voltages greater than 45kV supported on both steel towers and wood poles.

Two design approaches are covered in the standard, namely:

- i) an Empirical design using a deterministic approach
- ii) a General design using a probabilistic approach

EN50341 contains a series of National Normative Aspects (NNAs) in which each nation's National Committee presents design considerations which will apply in their country. In the UK NNA the User is given a choice of an Empirical or General design approach when considering an overhead line supported on wood poles. It was decided to adopt the Empirical design approach as this provided a design which was similar to the tried and tested method described in Issue 1.

Supports will generally be single wood poles but H pole supports for use in longer spans and heavier angles are included. Provision is made in this standard for a maximum span length of 200m. The design is based on the use of post type insulators of porcelain or polymeric material for intermediate positions and normal tension insulators of porcelain, glass or polymeric material for tension positions.

A clashing model has been added to the standard using a semi-probabilistic approach based on the method described in ENATR 111 as this has been shown to give good results in practice. Although the rest of the standard uses a deterministic approach with regard to weather loading the clashing model retains the regional and altitude specific wind and ice loading maps.

Issue 2 uses a flexible Microsoft Excel spreadsheet package in which the User can select their preferred conductor and weather loading without the need to refer to the large number of reference tables which would otherwise be required.

2. Definitions

For the purpose of this standard the following definitions shall apply:

Aeolian Vibration: Conductor oscillation produced by low velocity winds (in the region of 1-7ms⁻¹) blowing across the conductor.

Average Span: The arithmetic average of a number of spans in a line or a section of line.

Auxiliary Equipment: Equipment other than that included in the line design which may be erected on the supports.

Basic Span: The span length used in sag/tension calculations.

Conductor Clashing: The electrical contact between two dissimilar line conductors of an overhead line displaced from their normal position by environmental forces such that the conductors are likely to touch.

Conductor Creep: Permanent long term elongation of the conductor.

Conductor Downpull: The downward vertical load imposed by the conductor at a support resulting from the gradients to the adjacent pole tops.

Conductor Uplift: The upward vertical load imposed by the conductor at a support resulting from the gradients to the adjacent pole tops.

Crossarm Hamper: A term used to include all pole top crossarm components and fixings.

Equivalent Span: The equivalent span of a line, or section of line between rigid points, on which the conductors are free to move longitudinally at the supports, and in which the tension for any condition of load and temperature is therefore the same in all spans, is the span in which the tension variations are identical with those in the line or section of line. Its length is obtained by dividing the sum of the cubes of the individual spans by the length of the line or section and taking the square root of the whole.

i.e.

$$\text{Equivalent Span} = \sqrt{\frac{L_1^3 + L_2^3 + L_3^3 + \dots}{L_1 + L_2 + L_3 + \dots}}$$

Where L_1 , L_2 and L_3 are the individual spans of the section.

Everyday Tension: The design stress in an unloaded conductor at 5°C. This value is used in sag/tension calculations to limit harmful conductor vibrations.

Factor of Safety: The ratio, allowed for in the design of the overhead line, between the ultimate stress in a component and the safe permissible stress in it.

Freezing Point Tension: The design tension of an unloaded conductor at 0°C in still air.

Ice Coordinate: The intercept of the ice axis of the weather incidence load line (see ENATR 111) whose value is given for differing altitudes and UK locations in the associated weather maps.

Intermediate Support: A support in a straight run of a line on which the conductors are supported on line post insulators.

Maximum Conductor Pressure (MCP): The maximum transverse component of applied conductor load when subjected to a wind pressure.

Maximum Conductor Weight (MCW): The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.

Maximum Span: The maximum permitted length of any span for the conductor spacing adopted.

Maximum Working Tension (MWT): The absolute maximum conductor tension assessed at - 5.6°C with wind and ice loading.

Minimum Failing Load (MFL): The minimum applied mechanical load that will cause failure by breakage of a component.

Over-tensioning: Excess tension applied above normal tension at the time of erection to compensate for conductor creep.

Pre-tensioning: The tension treatment applied to a conductor for a short duration before final erection tension is established to remove a proportion of conductor creep.

Rated Breaking Strength (RBS): The rating given to a component (usually a conductor) defining the maximum mechanical load that the component can withstand without breaking.

Recommended Span: The average span length in any section to which the line should be planned.

Sag: The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span.

Section Angle Support: A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Section Support: A support in a straight run of a line in which the conductors are made off on either side of the crossarm on tension insulator sets.

Span: The horizontal distance between adjacent supports.

Wind Coordinate: The intercept on the wind pressure axis of the weather incidence load line (see ENATR 111) whose value is given for differing altitudes and UK locations in the associated weather maps.

Weather Zone: A geographical area in which the likely mean wind pressure and absolute maximum ice accretion thickness may be described by a number and a letter respectively.

The wind coordinate is described in 190N/m^2 increments whilst the ice coordinate is measured in 10mm diametric ice increments.

Weight Span: The sum of the distances, either side of a support, to the real or apparent invert points of the catenaries of the adjacent spans.

Wind Loading Span: The wind loading span associated with any support is half the sum of the spans adjacent to the support.

3 References

This standard makes reference to the following documents:

The Electricity Safety, Quality and Continuity Regulations 2002, SI 2665.

EN 50341 Overhead Electric Lines Exceeding AC 45kV.

BS 5950 Part 1: Structural Use of Steel in Building – Code of Practice.

BSEN ISO 898-1 Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel – Part 1: Bolts, Screws and Studs.

BS 3288 Insulator and Conductor Fittings for Overhead Power Lines.

BS1990 Part 1 Wood Poles for Overhead Lines (Power and Telecommunications Lines).

BS EN 14229 Structural Timber: Wood Poles for Overhead Lines..

EN 50182 Conductors for Overhead Lines – Round Wire Concentric Lay Stranded Conductors.

ENATS 43-88 Selection and Treatment of Wood Poles and Associated Timber for Overhead Lines.

ENATS 43-89 Auxiliary Equipment Erected on Overhead Line Supports

ENATS 43-90 Anti-Climbing Devices for HV Lines up to and Including 400kV.

ENATS 43-91 Stay Strands and Stay Fittings for Overhead Lines.

ENATS 43-92 Conductor Terminations, Joints and Insulator Binds for Overhead Lines up to and Including 132kV.

ENATS 43-93 Line Insulators.

ENATS 43-95 Steelwork for Overhead Lines.

ENATS 43-96 Fasteners and Washers for Wood Pole Overhead Lines.

ENATS 43-125 – Design Guide and Technical Specification for Overhead Lines Above 45kV

ENA Engineering Recommendation L38/2 Protective Greasing of PVC-Covered Copper and PVC Covered Aluminium Based Overhead Line Conductors.

4 Design Data

4.1 Weather Loadings

The weather loadings employed in this standard consist of combined wind and ice loadings in both a normal and an extreme environment. The wind and ice loadings in these environments are:

Normal environment: 380N/m^2 wind pressure and 9.5mm radial ice thickness.

Extreme environment: 570N/m^2 wind pressure and 12.5mm radial ice thickness.

A normal environment is defined as any environment in the United Kingdom (except Scotland) at an altitude not exceeding 300m. In Scotland a normal environment is defined as any location with an altitude not exceeding 200m. More onerous requirements may be specified to sites within this altitude range if local knowledge dictates this.

Extreme environments are defined as being any UK location at an altitude exceeding 300m (200m in Scotland) but not exceeding 500m. For lines to be constructed at altitudes greater than 500m a special consideration using local meteorological data should be made.

Ice loadings are calculated using the density of glaze ice (913kg/m^3).

4.2 Conductors

The minimum factor of safety on conductors shall be 2.0 on their calculated breaking loads. This limitation is known as Maximum Working Tension (MWT) and occurs at minimum conductor temperature (-5.6°C).

The design tensions of aluminium based conductors at 5°C (Everyday Tension) in still air shall not exceed 20% of their calculated breaking loads to reduce the risk of aeolian vibration. The design tension of copper based conductors shall not exceed 33% of their calculated breaking loads for similar reasons.

The conductor mass used in the calculations includes an allowance for grease as specified in Engineering Recommendation L38/2.

4.3 Clashing Span

The maximum possible span length whilst avoiding conductor clashing can be calculated using the Clashing Span spreadsheet. The spreadsheet is based on the method found in ENATR 111 and uses smoothed gust and lull wind pressures of 1.832 times and 0.546 times the mean wind pressures respectively. The spreadsheet employs a gust factor of 1.5 and the Freezing Point Tension (FPT) calculated in the Conductor S&T spreadsheet.

Weather zones used to calculate the maximum clashing span can be obtained from the supplied weather maps. The wind and ice loads are raised by a withstand factor of 1.1.

It is recommended that the minimum weather zone employed by the User should be 2B in normal environments and 3C in extreme environments.

4.4 Crossarm Assemblies

Crossarm assemblies are designed according to the philosophy outlined in BS 5950 Part 1. The steelwork is designed to a maximum stress of 172N/mm^2 , this being equivalent to the combination of a material partial factor of 0.64 and a climatic load factor of 2.5 on the yield stress of mild steel (275N/mm^2 for grade Fe430B) as per table 4.3.11(c)/GB.1 of BS EN 50341. The steelwork dimensions indicated on the drawings found in the steelwork spreadsheets are derived from those included in ENATS 43-95.

4.5 Supports

The stress generated in intermediate supports is a bending stress due to the wind load on the insulators and conductors. The load is assumed to be acting at a point 1.5m above the pole top. The stresses generated in stayed supports are crippling stresses produced by stay tension, crossarm hamper weight, conductor weight and conductor downpull.

The loading capacity of the supports is calculated using the following values for *Pinus Sylvestris* (supplied by the UK pole producers) taken from ENATS 43-88.

- 1) Mean ultimate extreme fibre stress 53.35N/mm^2
- 2) Average Modulus of Elasticity 10054N/mm^2

4.6 Insulators and Fittings

A default factor of safety of 3.0 is afforded on tension fittings and insulators. Post insulators used in this specification have historically had a cantilever strength of 24kN (MFL) at the point of conductor attachment. This value can be varied in the associated spreadsheets.

4.7 Foundations

The minimum factor of safety on foundations is 2.5. Foundation calculations are based on four different grades of ground conditions. These are:

- 1) Type 1 wet peaty soil $100\text{kN/m}^2/\text{m}$ of depth (H pole structures only)
- 2) Type 2 normal ground $200\text{kN/m}^2/\text{m}$ of depth
- 3) Type 3 well drained hard ground $300\text{kN/m}^2/\text{m}$ of depth
- 4) Type 4 hard stony ground $400\text{kN/m}^2/\text{m}$ of depth

The resistance of the structure to sinking under strut load is calculated in the S Pole Strut Load and H Pole Strut Load spreadsheets using the following assumed net soil bearing capacities taken from Table 12.1 of ENATS 43-125.

Wet Peaty Soil	100kN/m^2
Normal Clay Soil	200kN/m^2
Well Drained Hard Ground	300kN/m^2
Well Drained Hard Stony or Rocky Ground	500kN/m^2

4.8 Fasteners

Fasteners should be dimensioned in accordance with ENATS 43-96 with allowable stresses given in BSEN ISO 898-1. Allowable stresses are based on bolt ultimate tensile strength:

- i) Grade 4.6 Bolts 400N/mm^2
- ii) Grade 8.8 Bolts 800N/mm^2

These strengths should be reduced by the Factor of Safety (2.5) resulting in a UTS of 160N/mm^2 and 320N/mm^2 for grade 4.6 and 8.8 bolts respectively.

Bolt shear strength across the bolt threads is equal to 75% of the reduced UTS to take into account the reduced cross section at this point.

5. Materials

All components shall comply with the appropriate British Standard, BSEN or ENA Technical Specification or equivalent.

5.1 Anti Climbing Devices and Safety Signs

Anti climbing devices and safety signs shall comply with ENATS 43-90.

5.2 Conductors

Conductors shall be in accordance with BS EN 50182 in respect of aluminium conductors. Conductors are to be greased in accordance with ENA Engineering Recommendation L38/2 before use in the line. When performing calculations the weight of the grease should be included in the final conductor weight.

5.3 Wood Poles

Wood poles shall be fabricated in accordance with BS EN 14229 and ENATS 43-88 parts 1 and 2.

They shall be fabricated to drawings 435002-435005, all holes being drilled to template, the centre of the template being placed along the effective centre line of the pole. Pole markings shall be made on the same side of the pole as that on which the drilling templates have been used. Pole tops shall be evenly halved, the angle of the apex being 120° .

| Where H poles are required the limbs shall be suitably matched and marked.

5.4 Conductor Terminations, Joints and Other Fittings

Conductor terminations, joints and other fittings shall be in accordance with ENATS 43-92.

5.5 Fasteners and Washers

Fasteners and washers shall be in accordance with ENATS 43-96 and BSEN ISO 898-1.

5.6 Insulators and Associated Fittings

Insulators and associated fittings shall be in accordance with ENATS 43-93 and BS 3288.

5.7 Stays and Stay Fittings

Stays and stay fittings shall be in accordance with ENATS 43-91.

5.8 Crossarm Steelwork

Crossarm steelwork shall be generally in accordance with ENATS 43-95 and designed to BS 5950 Part 1.

6 Technical Requirements for the Erection of Lines

6.1 Span Lengths

The basic/recommended span for the chosen conductor can be varied by the User as required. The chosen span length can be checked for clashing performance using the Clashing Span spreadsheet.

6.2 Supports

Supports shall be single or H pole structures assembled in accordance with drawings 435006-435016.

6.3 Maximum Angles of Deviation

Line deviations are not permitted on post insulators. The maximum angles of line deviation can be determined from the Angle or Section Xarm – S Pole and Angle and Section Xarm – H Pole spreadsheets. It is recommended that deviations are limited to 30° and 75° on S pole and H pole section supports respectively.

In extreme situations when line deviations greater than 75° cannot be avoided the line should be terminated in each direction with terminal structures in close proximity with suitable jumpers in between.

6.4 Wind Loading Spans

Wind loading spans for intermediate supports are calculated by the User via the S Pole Windspans and H Pole Windspans spreadsheets. Users should enter the pole dimensions and the soil type and the windspans for various foundation arrangements are then returned. Tables of maximum / minimum pole sizes (taken from ENATS 43-88) are included for the User's guidance.

When considering intermediate structures the pole factor of safety shall be increased with increasing ground declination. These factors of safety are as follows:

- i) level ground to 1:10 declination FoS 2.5
- ii) 1:10 to 1:7.5 declination FoS 3.0

- iii) 1:7.5 to 1:5 declination FoS 3.5

The appropriate Factor of Safety for unstayed poles can be entered by the User into the labelled cell of the Front Page spreadsheet.

6.5 Vertical Conductor Loads and Weight Span

The vertical conductor load (V) imposed on a crossarm can be assumed, for practical purposes, to be the sum of the weight of conductor in half the adjacent spans plus the downpull or uplift imposed by the gradient on both sides of the support. This may be calculated as follows:

$$V(kg) = \frac{L_1 + L_2}{2} W + T_{\max} (\tan \theta_1 + \tan \theta_2)$$

$$WeightSpan(m) = \frac{L_1 + L_2}{2} + \frac{T_{\max}}{W} (\tan \theta_1 + \tan \theta_2)$$

Where

L_1 and L_2 are adjacent span lengths in metres

W is the weight of ice loaded conductor (augmented mass) in kg per metre

$T_{\max}$ is the maximum design tension for the conductor in kgf

$\tan \theta_1$ and $\tan \theta_2$ are the tangents of the gradients on each side of the support

For a downpull condition $\tan \theta$ is positive, conversely for an uplift condition $\tan \theta$ is negative.

6.6 Pilot Insulators

Where pilot insulators are erected on any structure, an equal number per phase shall be used in order to maintain a balanced potential at the crossarm.

In order to maintain electrical clearance it is recommended that armour rod type conductor clamps are fitted to these insulators.

6.7 Bonding

To avoid a capacitive induced potential at the point of attachment of insulator support steelwork to the crossarm it is important that the bases of all three insulators are electrically connected together. This is normally achieved by attachment to the crossarm but if for any reason it is necessary to attach an individual insulator directly to any part of a wood pole, the base of that insulator must be bonded to the bases of the insulators supporting the other two phases.

6.8 Provision of Danger Plates

Each support on a line to this standard shall be provided with a danger plate in accordance with ENATS 43-90. The plate shall be mounted approximately 3m above ground level, above any anti climbing guard and clearly visible to an observer on the ground.

6.9 Precautions against Access

The types of support requiring precautions against access and the methods to be employed are detailed in ENATS 43-90.

6.10 Minimum Height of Line Conductors and Clearances

The minimum height of line conductors shall comply with the Electricity Safety, Quality and Continuity Regulations 2002.

6.11 Relationship with Other Bodies

Where lines built to this standard are erected over or adjacent to the plant of Network Rail, Waterway Authorities or Telecommunications Operators then the relevant joint agreements shall be complied with.

6.12 Downleads

The wood pole terminal structure has the facility to accept downleads from a terminal tower or substation structure.

6.13 Line Termination on Steel Tower

Where required, the line may be terminated using steel towers as shown in drawing 435017. When a steel tower is used to terminate the line, the adjacent support shall have a conductor spacing of 2.5m.

6.14 Pole Mounted Cable Termination

A general arrangement for a pole mounted cable termination is given in drawings 435015 and 435016. All steelwork on this structure shall be bonded together and connected to earth in accordance with ENATS 43-89. Earthing arrangements for cable sealing ends and surge diverters shall be in accordance with the cable system requirements.

6.15 Downlead and Jumper Clearances

The following minimum clearances from downleads and jumpers must be maintained and, where they are free to move, such movement must be taken into account.

Phase to phase 1.5m

Phase to support steelwork or wood pole 1.2m

6.16 Staying Arrangements

Staying arrangements can be chosen from the User generated results given in the S Pole Stays, H Pole Stays and Terminal Stays spreadsheets as appropriate. Double stays are always to be splayed with the minimum spacing generated by the Stay Foundations spreadsheet in order to ensure that each stay has independent foundations. The magnitude of this angle is a function of the stay angle and the pole dimensions.

6.17 Support Strut Loading

The strut (vertical) load on the support is User calculated via the S Pole Strut Load and H Pole Strut Load spreadsheets. This is then compared to the maximum permissible strut load and the strut load as a percentage of the permissible strut load is returned. The strut loading of stayed supports can be minimised by using the largest possible stay angle.

6.18 Conductor Design/Erection Sags and Tensions

Ordinates for the plotting of uplift and ground clearance curves are generated by the User via the Conductor S&T spreadsheet and are displayed in a table. The conductor design and erection sags and tensions are generated in a similar manner.

7 Guidance for the Erection of Conductors

7.1 Conductor Creep

Erected conductors are subject to stresses which produce long term elongation known as conductor creep. The result of this is an increase in sag over the lifetime of the line and this must be anticipated and dealt with by the designer.

Conductor elongation due to creep consists of two components, namely:

- i) Metallurgical creep which is a function of the duration for which the load is applied
- ii) Geometrical settlement which is a function of the magnitude of the load and the stresses found at the conductor strand crossing points as the strands tighten and deform.

The degree of elongation varies between conductor types and is a function of both the erection tension and the mechanical and thermal conditions which occur during the lifetime of the line. The effect of conductor creep can be countered by use of the following techniques.

- i) Ordinate shift
- ii) Over-tensioning
- iii) Pre-tensioning

These techniques are described in the following sections.

7.2 Ordinate Shift

The ordinates for plotting ground clearance curves assume the use of a new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to a reduction in the design tension value at a temperature of 15°C.

7.3 Over-tensioning

The conductor can be over-tensioned at the time of erection, the resulting decrease in sag compensating for any conductor creep during the life of the line. The increase in conductor

tension can be expressed as a percentage increase over the design tension. A typical over-tension value is 10% at 15⁰C.

7.4 Pre-tensioning

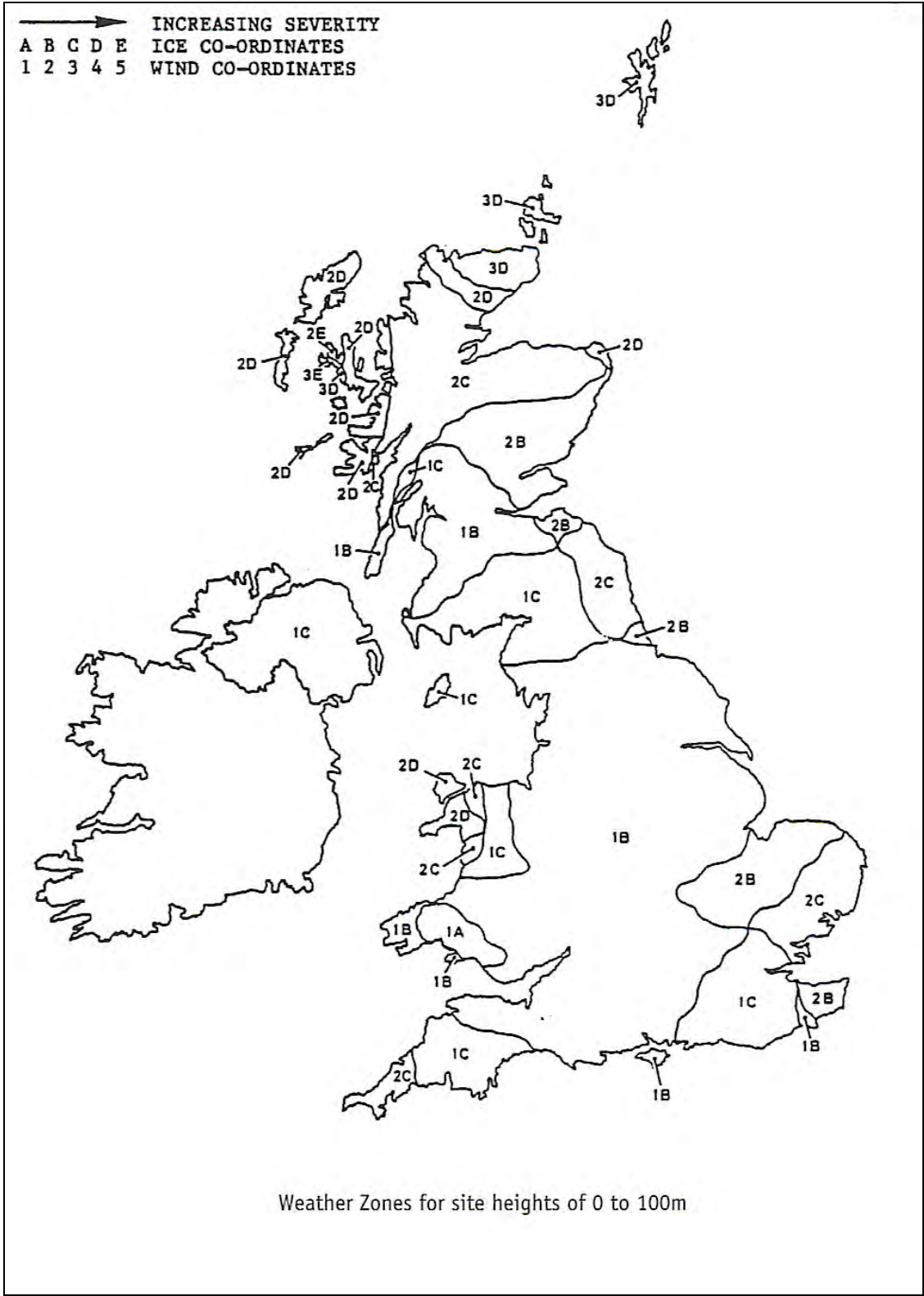
Before the final sagging of the conductor takes place it can be pre-tensioned in order to reduce the amount of elongation which will occur during the life of the line. The pre-tension value should not exceed either 50% of the conductor rated breaking strength (RBS) or the conductor maximum working tension (MWT).

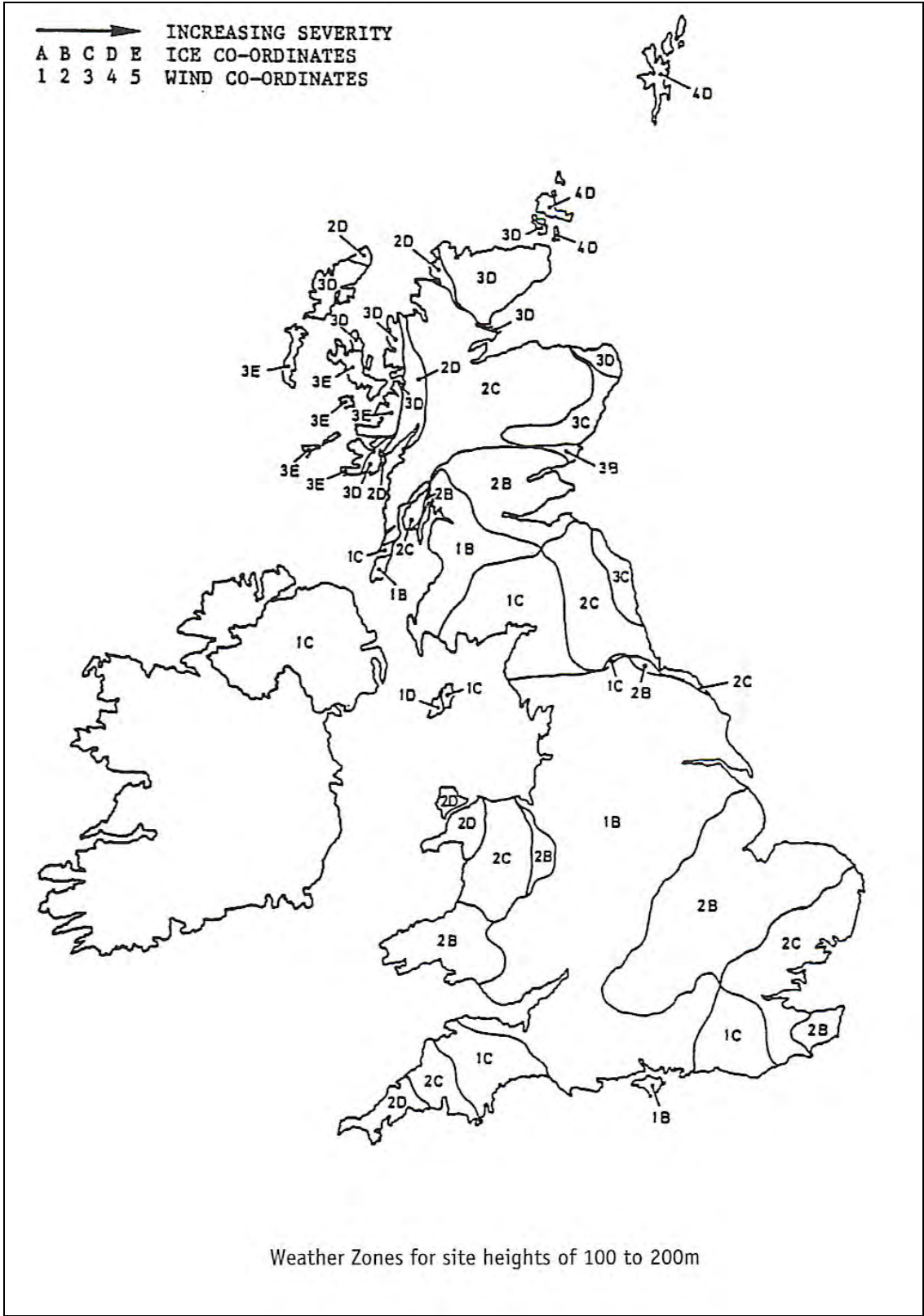
Conductors should be pre-tensioned to the required value for a minimum of 1 hour. The tension shall be adjusted to the pre-tension level at 15 minute intervals. At the end of this period the conductors shall be tensioned in accordance with the appropriate erection tension table and then terminated.

8 Weather Coordinate Maps for Conductor Clashing Assessment

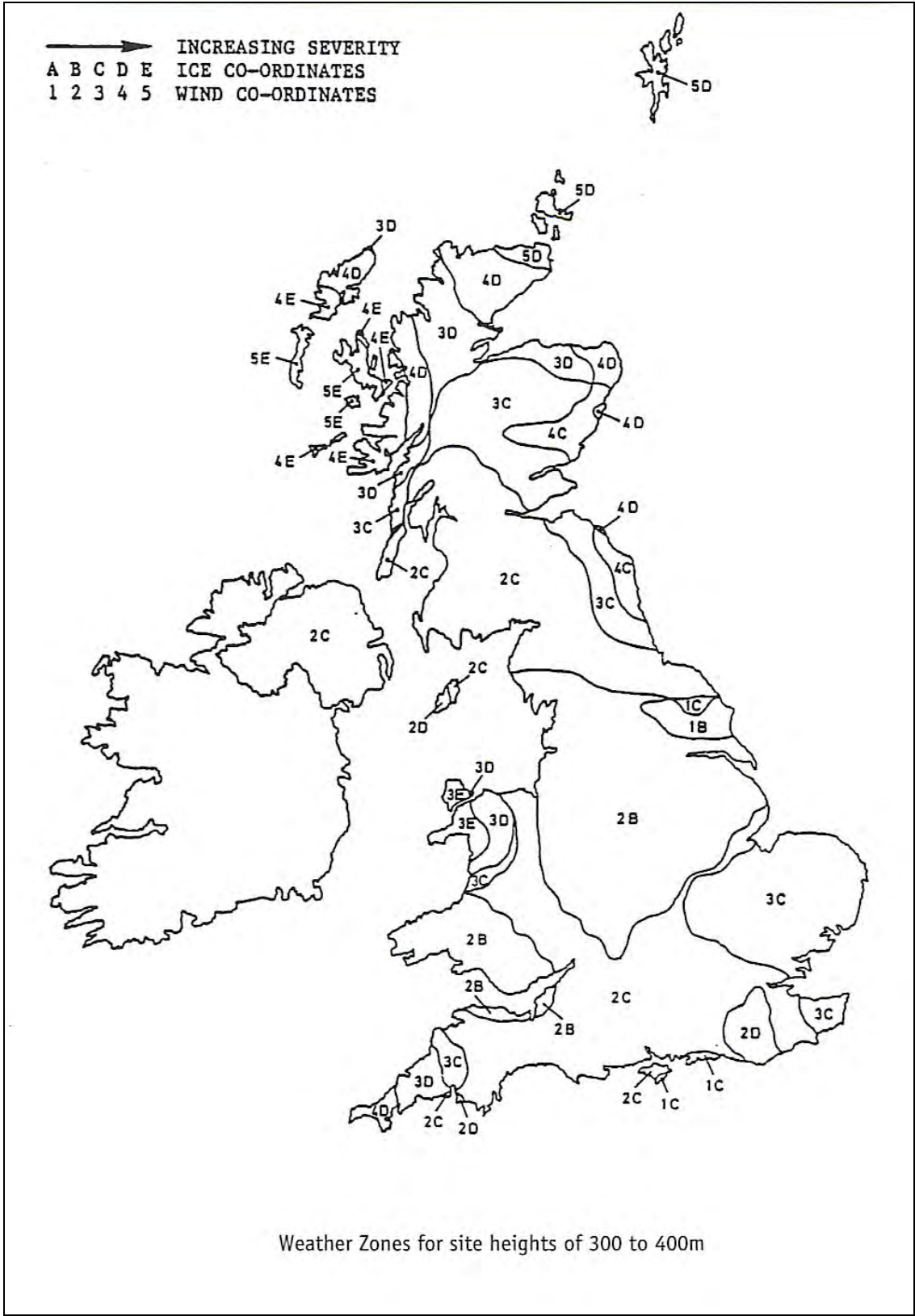
The clashing span spreadsheet uses the familiar ice and wind pressure coordinates in order to calculate the maximum permissible span length. These maps cover locations throughout the UK up to and including an altitude of 500m above sea level. Five maps are included covering the UK in 100m intervals.

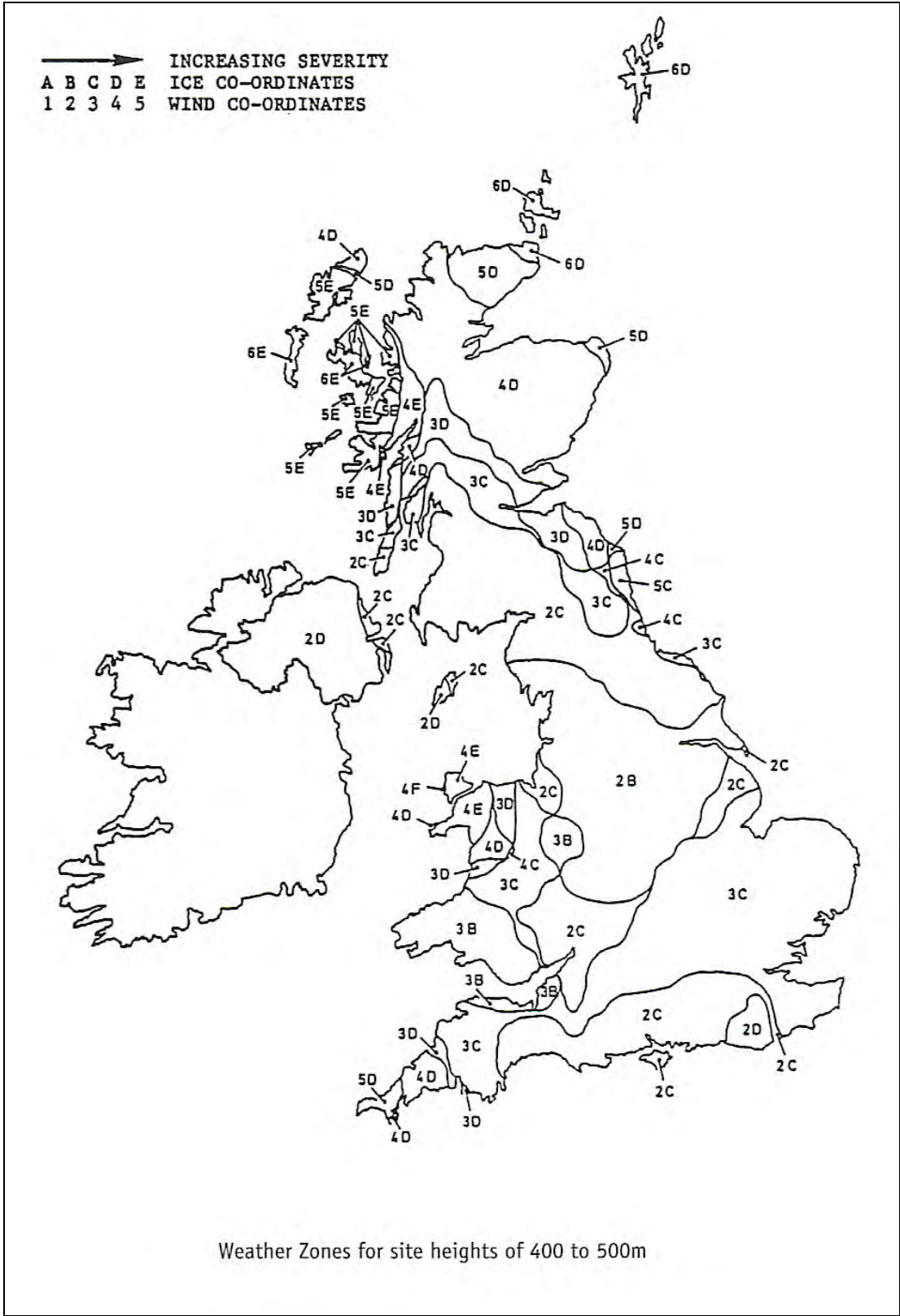
The maps are presented in the following pages.











9 General Assembly Drawings and Material Lists

This section presents drawings of the various structures as given in Issue 1 complete with a materials list. The following table contains a description of the various structures alongside the appropriate drawing number. It should be noted that these drawings are given as a general guide to each structure type.

Drawing Number and Title	Application of Structure
435001 132kV wood pole design – outline detail	N/A
435002 Single pole fabrication	All S pole structures
435003 Intermediate H pole fabrication	All intermediate H pole structures
435004 Terminal and section angle H pole fabrication	All terminal and section H pole structures
435005 Cable terminal structure auxiliary pole fabrication	Cable terminal structure
435006 General arrangement intermediate support – normal span single pole	Straight line – no deviation – S pole
435007 General arrangement intermediate support – long span single pole	Straight line – no deviation – S pole
435008 General arrangement intermediate support – long span H pole	Straight line – no deviation – H pole
435009 General arrangement angle / section support – Single pole boat crossarm	Light deviation – S pole
435010 General arrangement angle / section support – Single pole straight crossarm	Light deviation – S pole
435011 General arrangement angle / section support – H pole boat crossarm	Heavy deviation – H pole
435012 General arrangement angle / section support – H pole straight crossarm	Heavy deviation – H pole
435013 General arrangement terminal H pole – boat crossarm	Termination structure – connected to switchgear or an overhead line
435014 General arrangement terminal H pole – straight crossarm	Termination structure – connected to switchgear or an overhead line
435015 General arrangement pole mounted cable termination – boat crossarm	Termination structure with cable
435016 General arrangement pole mounted cable termination – straight crossarm	Termination structure with cable
435017 General arrangement steel tower termination structure	Steel tower termination
435018 Cable backboard	Wooden backboard for cable termination

10 Description of Spreadsheets

10.1 Introduction

Issue 1 of 43-50 presented information to the design engineer in the form of lookup tables thus eliminating the need for lengthy calculations. Additionally this issue only included information on three conductors (Lynx, Caracal and Jaguar).

Issue 2 has changed to a spreadsheet based format giving the User a greater degree of flexibility. The User may now choose either a pre-entered conductor type or choose a User Defined option and can also determine design parameters such as maximum windspans and angles of deviation by the application of weather conditions and factors of safety.

NOTE: It is recommended that a backup copy of the spreadsheet is made and saved without any alterations in case of problems or data corruption.

10.2 General Information

The design is based around an empirical approach as described in the UK NNA of EN50431.

The User can vary cell contents written in blue font. Cell contents written in red font are intended to remain unaltered. The worksheets are password protected and the majority of the cells are hidden from view. If the user requires access to the hidden cells the password to be used is "trident" (case sensitive). The hidden cell can be exposed by looking for the location where there is a discontinuity in the alphabetical increase in column names, highlighting the column on either side of the discontinuity, selecting format, column then unhide column.

Some cells contain comments providing relevant information. These cells are denoted by a red triangle in their top right hand corner. The comment can be read by running the mouse cursor over this triangle.

10.3 Front Page

The front page is used to enter some of the basic factors to be used in the line design. These include the conductor type, the weather data and the factors of safety for the components used in the construction.

The conductor can be chosen from the drop down list. This list contains a User Defined option, if this is chosen the User should enter the data into the fields found in the lower section of this spreadsheet.

The wind pressure can be chosen from 380N/m^2 (for normal environments) and 570N/m^2 (for extreme environments). Similarly the radial ice thickness can be chosen from 9.5mm (for normal environments) and 12.5mm (for extreme environments).

The weather zone can be chosen (for use in the clashing calculation only) in order to use the clashing model described in ENATR 111 and the UK NNA of EN50341.

Factors of safety are generally 2.5.. In order to compensate for the increased crippling load associated with increases in ground gradient three factors of safety for unstayed poles are to be used. These are:

- a) 2.5 for gradients up to and including 1:10
- b) 3.0 for gradients up to and including 1:7
- c) 3.5 for gradients up to and including 1:5.

Factors of safety are given in the UK NNA of EN50341

10.4 Conductor S&T

This spreadsheet is used to generate the sag and tension information for each conductor type. The User can vary the basic span around which the calculations are performed. All other data is automatically selected using the data entered into the **front page** sheet. The tables generated by this sheet are:

- a) ordinates for plotting uplift and ground clearance curves
- b) design sag tension
- c) pretension
- d) erection sag tension

The limiting tensions used in the S&T spreadsheet can be varied by the User. The majority of Empirically designed wood pole lines are built to a maximum working tension of the lesser of 23kN (based on the capacity of a 70kN tension insulator limited by a factor of safety of 3) or the RTS of the conductor limited by a factor of safety of 2. Under these well known conditions the MET limit can be set to equal the EDT limit and will therefore effectively be removed from the calculation process i.e. the calculation will be based on the MWT and EDT limits only. However, if the user decides to use a higher capacity tension insulator and therefore increases the MWT it is recommended that a suitable MET limit be employed in order to limit the tensions.

If erecting conductors at higher than standard tensions the User should also be aware on the increased load on the overhead line components. They should also consider:

- i) CDM implications of using higher tensions
- ii) Increased duty on crossarms and terminal and angle structures
- iii) Manual handling issues due to increased component sizes.

If required the User can enter and save data for additional conductors. To do this the User should follow these steps.

1. Select the Conductor S&T Spreadsheet.
2. Select the Review tab from the menu at the top of the page.
3. Select Unprotect spreadsheet. When prompted enter the password "trident" (case sensitive).
4. Look for the discontinuity in the alphabetical increase in column names. Select the column on either side of the discontinuity, right click and select Unhide. The complete contents of the spreadsheet will be exposed.

5. Scroll to column CF. Instructions on adding a new conductor are contained in the comment in cell **CI16**.
6. When the new conductor has been added select the columns which were previously hidden. Right click and select Hide. The contents of the spreadsheet will be re-hidden.
7. Re-protect the spreadsheet by selecting Protect spreadsheet from the review tab. You will be prompted to enter the password twice. It is recommended that the password "trident" is used in order to maintain consistency.

10.5 Adjacent Angles

This spreadsheet is intended to calculate the minimum conductor spacing at both supports and the midspan when considering angle structures. The User should enter the angle of deviation at each support and the spreadsheet will return the spacing data along with a recommended maximum span.

10.6 Clashing Span

This spreadsheet will return the maximum span which can be achieved before conductor clashing will occur. The User should enter the recommended span, the conductor horizontal spacing (if considering an angle structure the minimum of the three spacings returned by the **adjacent angles** sheet), the angle of the outer insulators to the vertical (if considering an intermediate structure) and the height of the insulator from which the conductor vertical offset is derived. If an angle / section structure is being considered 0 should be entered into the insulator height field.

10.7 Crossarm Section Properties

This spreadsheet will calculate the maximum permissible crossarm loads and these values are then passed onto the steelwork spreadsheets used for the various pole top configurations. The User should enter the steel ultimate tensile stress, the yield stress and the modulus of elasticity. The dimensions of the angle cross section should be entered into the relevant fields on the diagram. The dimensions of the various crossarms currently used can be found in EATS 43-95.

10.8 Crossarm Strut Section Properties

This spreadsheet calculates the maximum permissible compressive load for the crossarm strut used for S pole structures. The dimensions of the two strut sections used can be found in EATS 43-95. The User should enter the strut length (the length between the two bolt hole centres), the ultimate tensile stress, the yield stress and the modulus of elasticity.

10.9 Crossarm Channel Section Properties

This spreadsheet calculates the maximum permissible loads for the crossarm support channel. The User should enter the channel dimensions in the relevant cells in the diagram. Again the mechanical properties of the steel should be supplied. Dimensions can be found in EATS 43-95.

10.10 Insulator Mounting Plate

The User should enter the dimensions and material properties of the insulator mounting plate used to fix the post insulators to the crossarm.

10.11 Xarm Cross Brace Properties

The User should enter the dimensions and material properties of the crossarm cross braces used for the angle and terminal structures.

10.12 Stay Wishbone

The User should enter the cross-sectional dimensions and material properties of the wishbone used to attach the stay assembly to the crossarm.

10.13 Intermediate Crossarm for a Single Pole

This spreadsheet analyses the loads on the steelwork components and bolts used in the intermediate single pole structure. The User can cater for both the short type and long type structure by varying the crossarm dimensions. The User should check the correct crossarm section and strut dimensions are being considered by checking the relevant fields as found under the diagram. The User should supply the information as required by the blue cells. The results and any recommended actions are displayed in the blue box at the bottom of the sheet.

10.14 Intermediate Crossarm for an H Pole

This spreadsheet is similar to the **intermediate crossarm for an S Pole** sheet. Again, the User should ensure the correct crossarm section dimensions are being used by checking the cells immediately under the diagram.

10.15 Angle/Sect Boat Xarm - S Pole

Again, the User should enter the crossarm dimensions into the relevant cells and ensure the correct crossarm and crossarm strut dimensions are being used. In addition to the information required by the previous two spreadsheets information concerning the stay bolts and angles, information concerning the line deviation and tension insulators should be provided.

10.16 Angle/Sect Boat Xarm - H Pole

Similar to above but the user should specify the number of stays to be used. This is to calculate the crippling load produced by each stay. The assumptions in this regard are:

- a) one stay – 100% on outer stay assembly
- b) two stays – 100% on outer stay assembly
- c) three stays – 40% on outer stay assembly, 60% on inner stay assembly
- d) four stays – 50% on each stay assembly.

10.17 Terminal Boat Xarm - H Pole

This spreadsheet offers the User a number of options. As described above the User should check the dimensions of the crossarm cross section and enter the steelwork dimensions in the relevant cells on the figure. The User is then given options as to the downlead connections. The User can select between a downlead connection to an overhead line or an item of switchgear. The configuration of the downleads (angles etc) can be entered into the minor figures below the main diagram. The User can then select whether or not sidestays are used to counter any transverse pull of the downlead connections. It is assumed that four backstays are always used with this crossarm assembly.

10.18 Angle/Sect Str Xarm – S Pole

As 10.15 but with straight crossarm sections.

10.19 Angle/Sect Str Xarm – H Pole

As 10.16 but with straight crossarm sections.

10.20 Terminal Str Xarm – H Pole

As 10.17 but with straight crossarm sections.

10.21 Conductor Strap

This spreadsheet checks the capacity of the conductor strap. The spreadsheet caters for both long and short type straps as the dimensions between the crossarm / strap bolt and the conductor attachment are the same in both instances. Again, the User should enter the mechanical properties of the steel used and enter the strap dimensions into the relevant cells on the diagram.

10.22 S Pole Windspans

This spreadsheet calculates the maximum windspan capability of both the pole and its associated foundations. The User should select the pole modulus of rupture and modulus of elasticity. The User should then enter the post insulator height and width. Finally the User should select a soil category and the distance from the groundline to the upper baulk bolt.

The soil categories are (as per ENATS 43-125):

- a) category 1 – wet peaty $100\text{kN/m}^2/\text{m}$
- b) category 2 – normal (clay) $200\text{kN/m}^2/\text{m}$
- c) category 3 – hard drained $300\text{kN/m}^2/\text{m}$
- d) category 4 – hard stony $400\text{kN/m}^2/\text{m}$

The windspans returned are as limited by the pole (stress at the critical diameter) and as limited by various foundation baulk configurations. These configurations are as shown in Issue 1 and are described in the comments in the relevant results box at the bottom of the sheet.

The suitability of the foundation baulks is also determined based on the bending moment produced in the block (derived from the block length and the ultimate fibre stress. If the baulk is too long the message "Baulk too Long" is generated and it should be replaced by the Type 1 (1500mm) version. If the baulk length is satisfactory the message "Baulk Length Suitable" is generated.

10.23 H Pole Windspans

This spreadsheet performs similar calculations to the above sheet but for H pole structures.

Foundation capabilities are given for no lateral baulks, type 1 lateral baulks (1500mm long) and type 2 lateral baulks (2500mm long).

10.24 S Pole Strut Loads

This spreadsheet calculates the crippling load in the S poles and the bearing loads on the supporting soil. Insulator data should be entered into the relevant cells. The spreadsheet assumes the steelwork weight is that of an S pole angle/section crossarm assembly. The User can use their own User Defined criteria or use the tables provided in the lower section of the worksheet.

The soil categories are (as per ENATS 43-125):

- a) category 1 – wet peaty 100kN/m²
- b) category 2 – normal (clay) 200kN/m²
- c) category 3 – hard drained 300kN/m²
- d) category 4 – hard stony 500kN/m²

10.25 H Pole Strut Loads

This spreadsheet is similar to the S pole sheet. The User should read the result from the relevant results box. This is described further in the comments included in the results cells.

The soil categories are (as per ENATS 43-125):

- a) category 1 – wet peaty 100kN/m²
- b) category 2 – normal (clay) 200kN/m²
- c) category 3 – hard drained 300kN/m²
- d) category 4 – hard stony 500kN/m²

10.26 Terminal Strut Loads

This spreadsheet is similar to those described in 10.23 and 10.24.

10.27 S Pole Stays

This spreadsheet calculates the maximum capacities of various stay configurations allowing the User to select one suitable for their needs. A table of stay spreads at common stay angles is given alongside a cell in which the user can enter their own stay angle. The pole dimensions and point of stay attachment to the crossarm can be entered by the User.

The User can select various stay wire grades and sizes. The maximum horizontal pole top load this stay can take is then returned and can be compared with the value of pole top load returned by the various steelwork spreadsheets. The maximum value of tension in each stay is also returned.

Indicative maximum angles of deviation taking into account the conductor loading and stay capacity are returned in the tables in the lower section of the spreadsheet.

10.28 H Pole Stays

Similar to the above. This spreadsheet allows the User to obtain values of maximum horizontal pole top load for either one, two three or four stays.

10.29 Terminal Stays

Similar to the above. This spreadsheet allows the User to obtain values of maximum horizontal longitudinal load produced by the conductor tension in the final span of the line. It is assumed four stays are always used in this configuration.

10.30 Stay Foundations

This spreadsheet allows the User to evaluate the capacity of the stay foundations. The User should enter the stay block dimensions, stay angle, stay rod length, stay block density and soil density. The stay wire properties should also be provided. The maximum stay tension, this tension as a percentage of the soil capability and the minimum spread to ensure each stay block has an individual frustum of soil is then returned in the results box.